“IPL Data Analysis: Exploring Team Performance, Match Trends and Key Insights”

A PROJECT SUBMITTED TO

SYNTECXHUB Technologies

Kanpur Nagar, Uttar Pradesh

FOR THE INTERNSHIP

IN

DATA SCIENCE

By

Karne Sayali Avinash

30-04-2026

Introduction

The Indian Premier League (IPL) is one of the most prominent and widely followed T20 cricket leagues in the world. Since its inception in 2008, the IPL has revolutionized the game of cricket by combining sports, entertainment and business on a global platform. With participation from top international and domestic players, the league has gained immense popularity and has become a significant event in the global sporting calendar.

Over the years, the IPL has generated a large volume of data, including match results, team performances, player statistics and venue details. This data provides an excellent opportunity for data analysis, enabling the extraction of valuable insights related to team strategies, match outcomes and tournament trends. Analyzing such data can help in understanding patterns that are not immediately visible and can support data driven decision making.

This project focuses on performing an in depth analysis of IPL match data to explore various aspects of the tournament. The study examines key factors such as team performance, toss decisions, match outcomes, venue distribution and season-wise trends. By applying data cleaning, exploratory data analysis (EDA) and visualization techniques, the project aims to identify meaningful relationships and trends within the dataset.

The primary objective of this analysis is to understand how different factors influence match results and to evaluate the consistency and performance of teams over multiple seasons. Additionally, the project seeks to analyze how the IPL has evolved over time in terms of the number of matches and the expansion of the tournament.

Through this analysis, the project provides a comprehensive overview of the IPL, highlighting key insights such as dominant teams, the impact of toss decisions and the importance of specific venues. The findings of this study can be useful for cricket analysts, enthusiasts and stakeholders interested in understanding the dynamics of the IPL from a data driven perspective.

Objectives

- To load and explore the IPL dataset using Pandas.
- To perform necessary data cleaning such as handling missing values and removing duplicates.
- To Explore the dataset to understand key patterns, distributions, and relationships between important variables.
- To identify the top performing teams based on the number of matches won.
- To Understand whether teams prefer to bat or field after winning the toss and identify common strategies.
- To Analyze which venues host the most matches and observe patterns related to match locations.
- To analyze the relationships and correlations between different match factors such as toss result and match outcome.
- To analyze the trend of matches played across different IPL seasons.

Methodology

The analysis was conducted using a structured data science approach, consisting of multiple stages from data collection to insight generation.

1. Data Collection

The dataset used for this project was obtained from publicly available sources (Kaggle). It contains IPL match level data, including information about teams, match results, toss outcomes, venues and seasons. The dataset represents historical IPL matches played across multiple seasons, making it suitable for trend and performance analysis.

2. Data Description

The dataset includes key features such as team names (team1, team2), match winner, toss winner, toss decision (bat/field), venue and season/year of the match. These attributes help in analyzing team performance, match strategies and seasonal patterns in the IPL.

3. Data Cleaning

The dataset was preprocessed by handling missing values, removing duplicate records and ensuring consistency in team names and other categorical variables. Appropriate techniques were used to handle both categorical and numerical data.

4. Exploratory Data Analysis (EDA)

Initial exploration was performed to understand the structure and distribution of the dataset. Key variables were analyzed to identify trends, patterns and relationships within the data.

5. Data Visualization

Various visualization techniques such as bar plots, line charts, heatmap and comparative graphs were used to represent findings effectively. These visualizations helped in better understanding team performance, toss impact, venue distribution and seasonal trends.

6. Insight Generation

Based on the analysis, meaningful insights were derived regarding team dominance, match trends and the influence of different factors on match outcomes. The final conclusions were drawn based on observed data patterns.

Data Analysis

- ❖ **To load and explore the IPL dataset using Pandas.**

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_csv("C:\\Users\\Dell\\Downloads\\ipl match.csv")


df.head()
```

Output:

Data columns (total 5 rows)

	id	season	city	date	match_type	player_of_match	venue
0	335982	2007/08	Bangalore	4/18/2008	League	BB McCullum	M Chinnaswamy Stadium
1	335983	2007/08	Chandigarh	4/19/2008	League	MEK Hussey	Punjab Cricket Association Stadium, Mohali
2	335984	2007/08	Delhi	4/19/2008	League	MF Maharoof	Feroz Shah Kotla
3	335985	2007/08	Mumbai	4/20/2008	League	MV Boucher	Wankhede Stadium
4	335986	2007/08	Kolkata	4/20/2008	League	DJ Hussey	Eden Gardens

	team1	team2	toss_winner	toss_decision	winner
0	Royal Challengers Bangalore	Kolkata Knight Riders	Royal Challengers Bangalore	Field	Kolkata Knight Riders
1	Kings XI Punjab	Chennai Super Kings	Chennai Super Kings	Bat	Chennai Super Kings
2	Delhi Daredevils	Rajasthan Royals	Rajasthan Royals	bat	Delhi Daredevils
3	Mumbai Indians	Royal Challengers Bangalore	Mumbai Indians	Bat	Royal Challengers Bangalore
4	Kolkata Knight Riders	Deccan Chargers	Deccan Chargers	Bat	Kolkata Knight Riders

	result	result_margin	target_runs	target_overs	umpire1	umpire2
0	Runs	140	223	20	Asad Rauf	RE Koertzen
1	Runs	33	241	20	MR Benson	SL Shastri
2	Wickets	9	130	20	Aleem Dar	GA Pratapkumar
3	Wickets	5	166	20	SJ Davis	DJ Harper
4	Wickets	5	111	20	BF Bowden	K Hariharan

- **To perform necessary data cleaning such as handling missing values and removing duplicates.**

Code:

```
import pandas as pd


# Load dataset
df = pd.read_csv("C:\\Users\\Dell\\Downloads\\ipl match.csv")


# Basic info
print(df.info())
print(df.isnull().sum())


# Remove duplicates
df.drop_duplicates(inplace=True)


# Handle missing values properly
cat_cols = df.select_dtypes(include='object').columns
num_cols = df.select_dtypes(include=['int64', 'float64']).columns


df[cat_cols] = df[cat_cols].fillna("Unknown")
df[num_cols] = df[num_cols].fillna(0)
```

Output:

RangeIndex: 1095 entries, 0 to 1094

Data columns (total 19 rows):

	Column	Non-Null Count	Dtype
0	id	1095 non-null	int64
1	season	1095 non-null	Object
2	city	1044 non-null	Object
3	date	1095 non-null	Object
4	match_type	1095 non-null	Object
5	player_of_match	1090 non-null	Object

6	venue	1095 non-null	Object
7	team1	1095 non-null	Object
8	team2	1095 non-null	Object
9	toss_winner	1095 non-null	Object
10	toss_decision	1095 non-null	Object
11	winner	1090 non-null	Object
12	result	1095 non-null	Object
13	result_margin	1076 non-null	float64
14	target_runs	1092 non-null	float64
15	target_overs	1092 non-null	float64
16	super_over	1095 non-null	Object
17	umpire1	1095 non-null	Object
18	umpire2	1095 non-null	Object

dtypes: float64(3), int64(1), object(15)

memory usage: 162.7+ KB

None

Column	Non-Null Count
id	0
season	0
city	0
date	0
match_type	0
player_of_match	0
venue	0
team1	0
team2	0
toss_winner	0
toss_decision	0
winner	0
result	0
result_margin	0
target_runs	0
target_overs	0
super_over	0
umpire1	0
umpire2	0

Interpretation

The dataset was cleaned by converting data into appropriate formats and handling missing values. The date column was properly formatted for analysis. Now the dataset is clean, structured and ready for further analysis.

❖ **To Explore the dataset to understand key patterns, distributions and relationships between important variables.**

Code:

```
# Describe dataset
print(df.describe())

# Unique teams
print(df['team1'].unique())

# Match count
print("Total Matches:", len(df))
```

Output:

	id	result_margin	target_runs	target_overs
count	1.095000e+03	1095.000000	1095.000000	1095.000000
mean	9.048283e+05	16.959817	165.230137	19.705205
std	3.677402e+05	21.714792	34.487313	1.887000
min	3.359820e+05	0.000000	0.000000	0.000000
25%	5.483315e+05	5.000000	146.000000	20.000000
50%	9.809610e+05	8.000000	166.000000	20.000000
75%	1.254062e+06	19.000000	187.000000	20.000000
max	1.426312e+06	146.000000	288.000000	20.000000

❖ **To identify the top performing teams based on the number of matches won.**

Code:

```
import matplotlib.pyplot as plt

team_wins = df['winner'].value_counts()

print(team_wins)

import seaborn as sns
```

```
sns.countplot(y='winner', data=df, order=df['winner'].value_counts().index)
plt.title("Match Wins by Team")
plt.show()
```

Output:

Winner

Mumbai Indians	144
Chennai Super Kings	138
Kolkata Knight Riders	131
Royal Challengers Bangalore	116
Rajasthan Royals	112
Kings XI Punjab	88
Sunrisers Hyderabad	88
Delhi Daredevils	67
Delhi Capitals	48
Deccan Chargers	29
Gujarat Titans	28
Punjab Kings	24
Lucknow Super Giants	24
Gujarat Lions	13
Pune Warriors	12
Rising Pune Supergiant	10
Royal Challengers Bengaluru	7
Kochi Tuskers Kerala	6
Rising Pune Supergiants	5

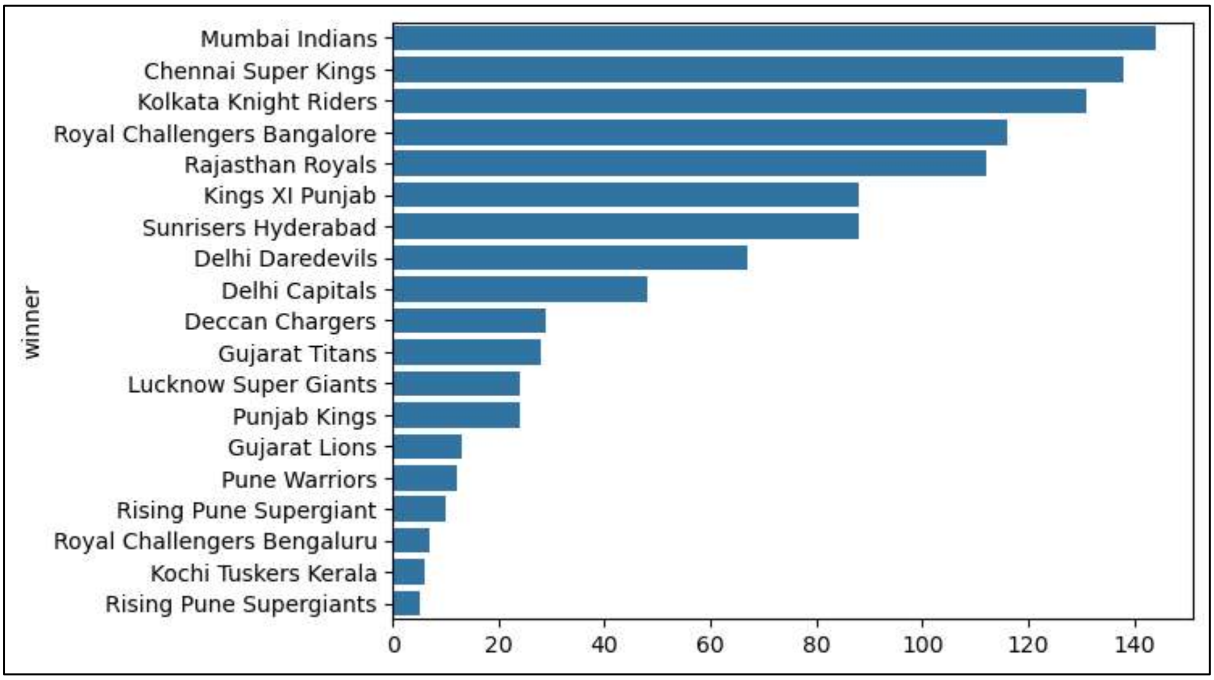

Fig.1

Interpretation

- The analysis shows that **Mumbai Indians** have the highest number of wins **(144)**, making them the most successful team in IPL history.
- **Chennai Super Kings (138)** and **Kolkata Knight Riders (131)** are also among the top-performing teams, indicating consistent performance across seasons.
- **Royal Challengers Bangalore** and **Rajasthan Royals** have also performed well, with over 100 wins each.
- Mid level teams like **Kings XI Punjab** and **Sunrisers Hyderabad** have moderate performance with around 80–90 wins.
- Teams such as **Deccan Chargers**, **Gujarat Titans** and **Lucknow Super Giants** have fewer wins, possibly due to fewer seasons played.
- Some teams like **Kochi Tuskers Kerala** and **Pune Warriors** have very low win counts, indicating limited participation or shorter team lifespan.

Mumbai Indians dominate the IPL in terms of total wins, followed closely by Chennai Super Kings and Kolkata Knight Riders.

Conclusion

The analysis of IPL match data reveals that team performance varies significantly across franchises. Mumbai Indians emerge as the most dominant team, followed by Chennai Super Kings and Kolkata Knight Riders, showcasing consistent success over multiple seasons. While some teams have maintained strong performance, others have lower win counts due to fewer appearances or inconsistent results. Overall, the data highlights the competitive nature of the IPL and the importance of consistency in achieving long term success.

❖ **To Understand whether teams prefer to bat or field after winning the toss and identify common strategies.**

Code:

```
toss_decision = df['toss_decision'].value_counts()


print(toss_decision)


toss_decision.plot(kind='bar')

plt.title("Toss Decision (Bat vs Field)")

plt.xlabel("Decision")

plt.ylabel("Count")
```

```
plt.show()
```

Output:

toss_decision

field : 704

bat : 391

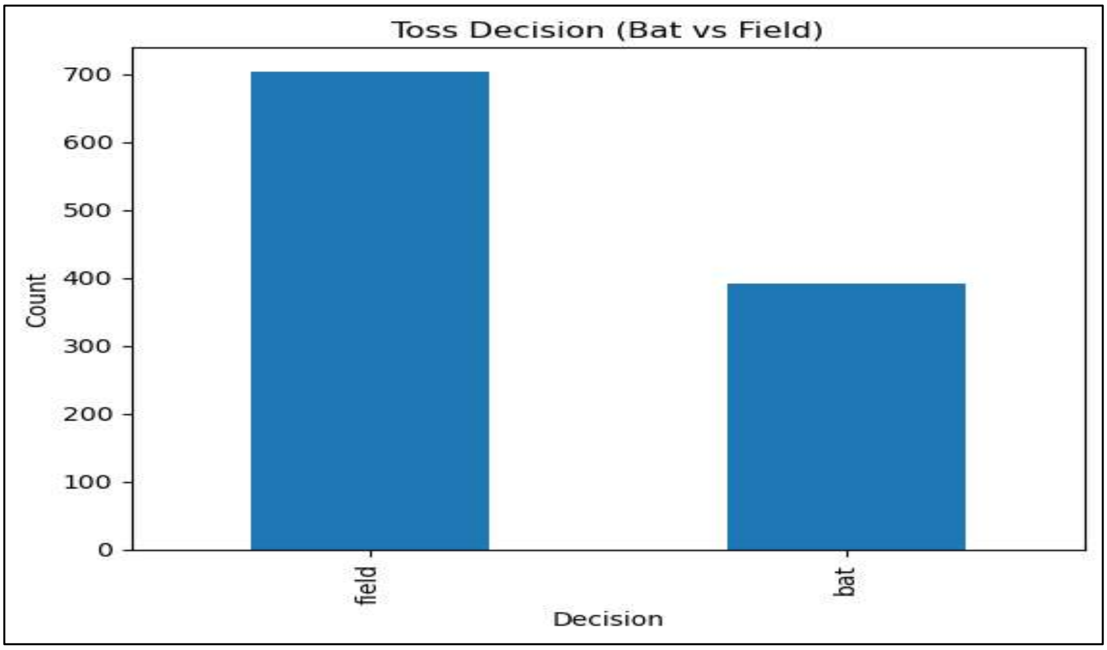

Fig.2

Interpretation

The analysis shows that most teams prefer to choose fielding after winning the toss, as the number of field decisions is significantly higher than batting decisions.

Conclusion

Teams in the IPL generally prefer to field first, indicating a common strategy that may be influenced by factors such as pitch conditions, dew and chasing advantage.

- **To Analyze which venues host the most matches and observe patterns related to match locations.**

Code:

```
venue = df['venue'].value_counts().head(10)

print(venue)


venue.plot(kind='bar')
```

```
plt.title("Top Venues by Matches")
plt.xlabel("Venue")
plt.ylabel("Number of Matches")
plt.xticks(rotation=90)
plt.show()
```

Output:

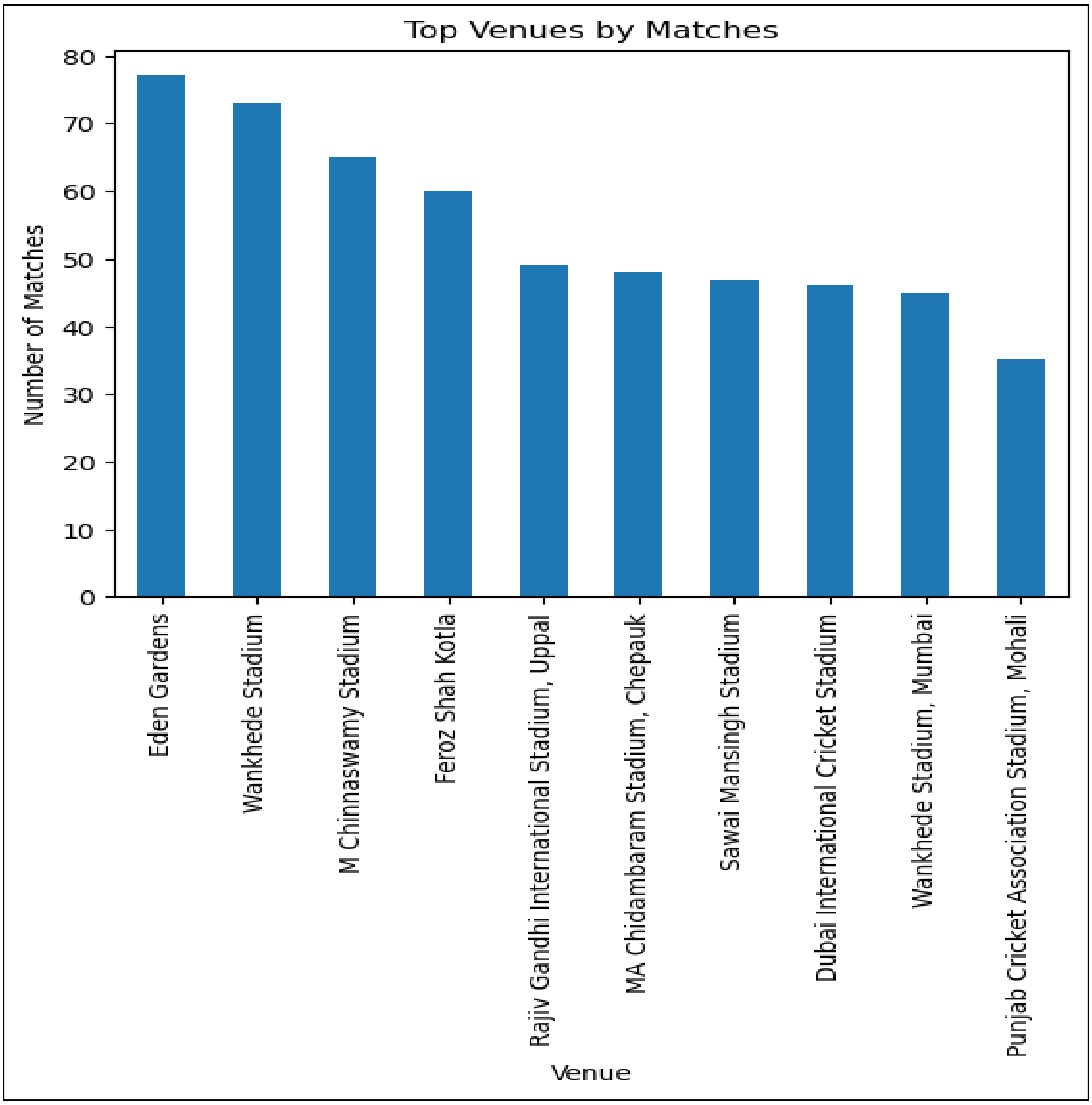

Fig.3

Interpretation

The analysis shows that Eden Gardens hosts the highest number of matches, followed by Wankhede Stadium and M. Chinnaswamy Stadium. These venues are key locations in the IPL, while other stadiums host comparatively fewer matches.

Conclusion

The distribution of matches across venues is uneven, with a few major stadiums hosting a large number of games. This highlights the importance of certain locations in the IPL, making them central hubs for hosting matches.

- ❖ **To analyze the relationships and correlations between different match factors such as toss result and match outcome**.

Code:

```
import seaborn as sns

import matplotlib.pyplot as plt


# Convert categorical to numeric (for heatmap)

df_encoded = df.copy()

df_encoded['toss_win_match_win'] = (df['toss_winner'] == df['winner']).astype(int)


sns.heatmap(df_encoded.corr(numeric_only=True), annot=True)

plt.title("Correlation Heatmap")

plt.show()
```

Output:

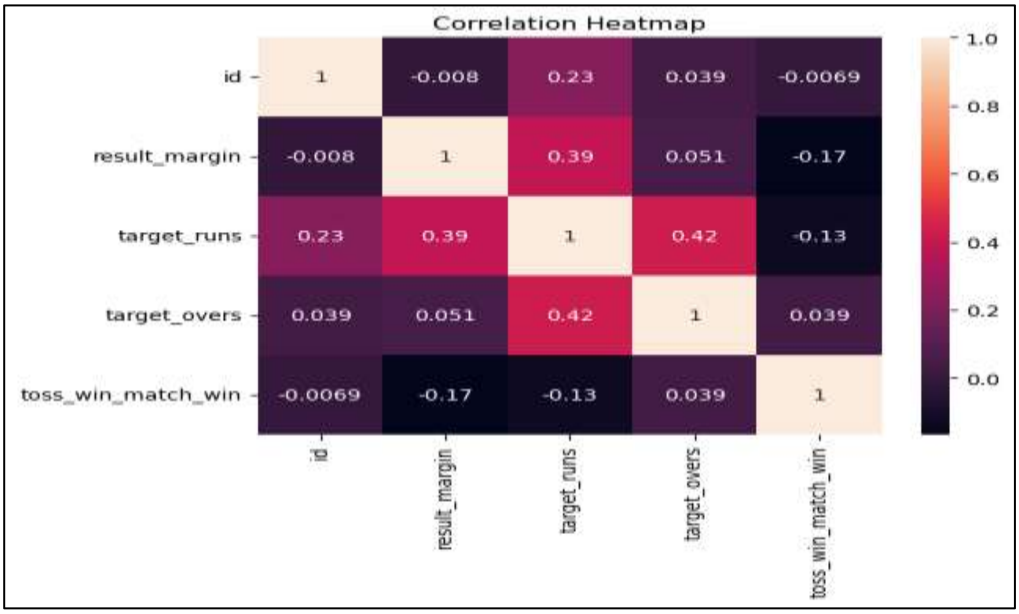

Fig.4

Interpretation

The heatmap shows that most variables have weak correlations with each other. A moderate positive relationship is observed between target runs and target overs, while toss outcome has very little impact on match results.

Match outcomes are not strongly dependent on toss results or individual numeric factors.

Conclusion

The analysis of IPL data highlights that team performance, match trends, and strategic factors such as toss decisions play an important role in the game. While teams like Mumbai Indians and Chennai Super Kings have shown consistent dominance, the overall number of matches has increased over the years, reflecting the growth of the IPL. Additionally, factors like toss results have only a limited impact on match outcomes. Overall, the IPL demonstrates a competitive and evolving structure with changing dynamics across seasons.

- **To analyze the trend of matches played across different IPL seasons.**

Code:

```
import matplotlib.pyplot as plt

import seaborn as sns


# Count matches per season (sorted)

season_matches = df['season'].value_counts().sort_index()


plt.figure(figsize=(10,5))  # increase size


sns.lineplot(x=season_matches.index, y=season_matches.values, marker='o')


plt.title("Matches Played per Season")

plt.xlabel("Season")

plt.ylabel("Matches")


# Fix x-axis labels

plt.xticks(season_matches.index, rotation=45)


plt.grid(True)
```

```
plt.tight_layout()

plt.show()
```

Output:

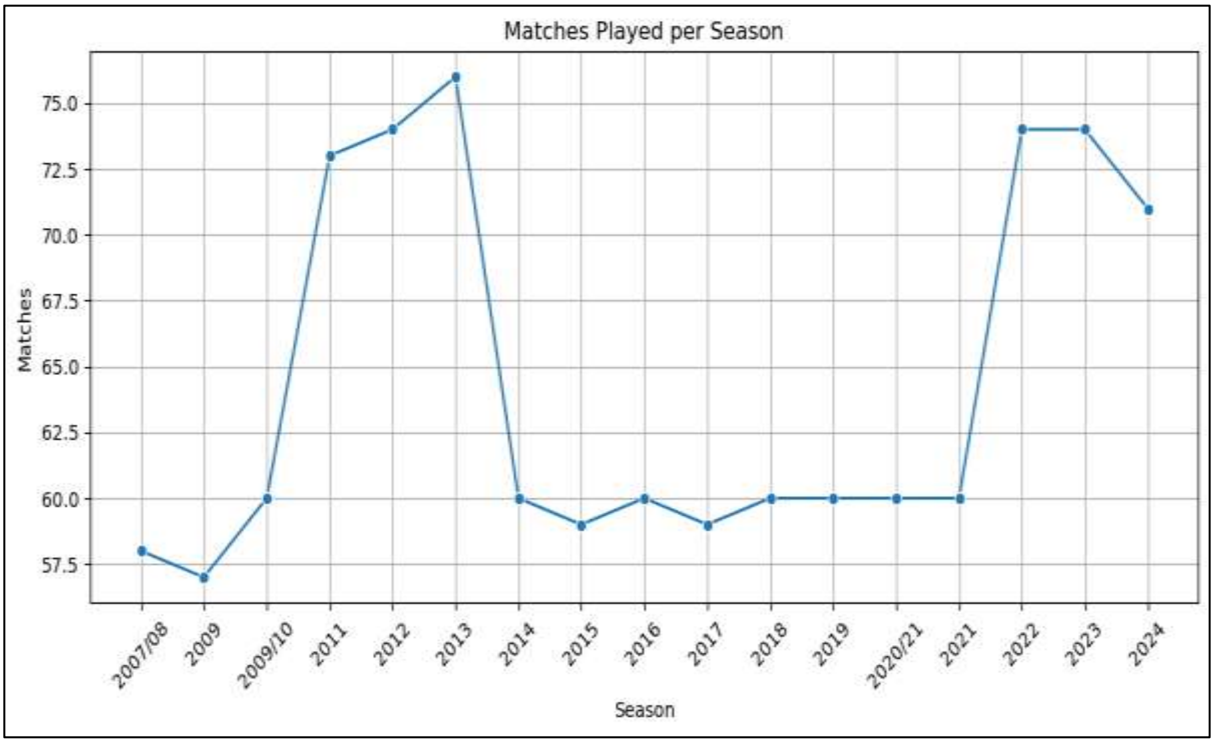

Fig.5

Interpretation

- The number of matches was relatively lower in the initial seasons (around 56–60 matches between 2008–2010).
- A significant increase is observed from 2011 to 2013, reaching a peak of around 76 matches in 2013, indicating expansion in the tournament format.
- There is a noticeable drop in 2014, followed by a stable phase from 2015 to 2021, where the number of matches remained consistent at around 59–60 matches per season.
- From 2022 onwards, the number of matches again increased sharply to around 74 matches, reflecting the inclusion of new teams and expansion of the league.
- A slight decline is observed in 2024, but the number of matches still remains higher than earlier seasons.

The IPL has shown overall growth over the years, with significant expansions leading to an increase in the number of matches in recent seasons.

Conclusion

The season-wise analysis indicates that the IPL has evolved significantly over time. While the initial seasons had fewer matches, the league expanded in later years, leading to an increase in the number of matches played. After a period of stability, recent seasons show renewed growth, highlighting the increasing scale and popularity of the tournament.

Overall Conclusion

This project provides a comprehensive analysis of IPL match data, focusing on team performance, match trends and key influencing factors. The study reveals that certain teams, such as Mumbai Indians and Chennai Super Kings, have consistently performed well over the years, highlighting their dominance in the league.

The analysis also shows that while factors like winning the toss may provide a slight advantage, they do not guarantee match success, indicating that overall team performance plays a more significant role. Additionally, the season-wise trends demonstrate the growth of the IPL, with an increase in the number of matches in recent years due to the expansion of the tournament.

Venue based analysis highlights that a few stadiums host a majority of matches, making them important centers for IPL events. Furthermore, the correlation analysis indicates that match outcomes are not strongly dependent on any single factor, emphasizing the competitive and unpredictable nature of the game.

Overall, the IPL is a dynamic and evolving tournament where multiple factors contribute to match outcomes. This analysis helps in understanding key patterns and provides valuable insights into the structure and growth of the league.